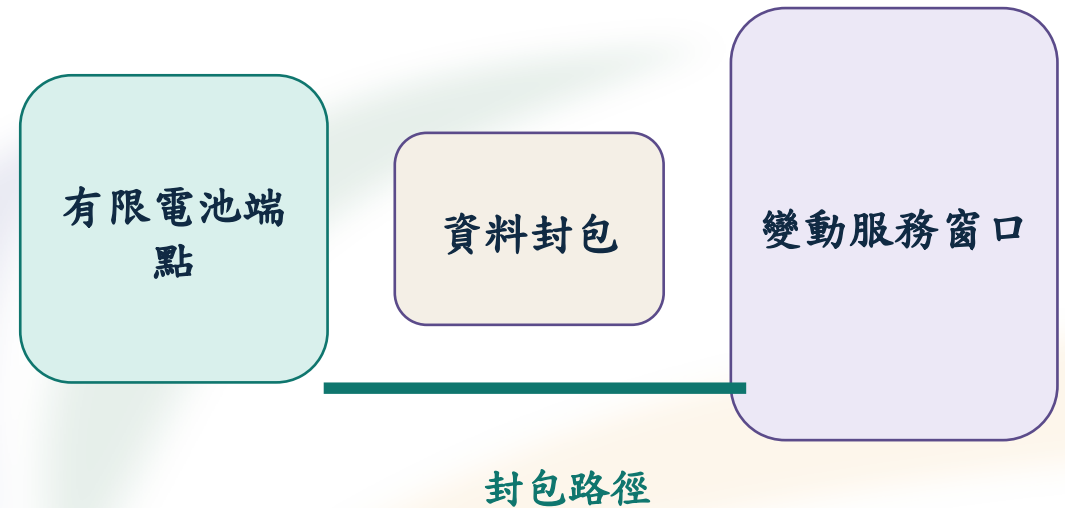


P001 | 有限電池與變動服務窗口

- *scenario*（情境定義）：source = course package；purpose = 固定端點、窗口與資料規則；unit = identifier；interpretation = 同一組輸入供各 case 重跑。
- 環境感測端點在固定 scenario 內產生資料封包；端點電池有限，服務窗口可開啟、關閉或縮短。
- *endpoint*（端點）：source = scenario fixture；purpose = 執行 policy；unit = classification；interpretation = 以端點 radio / processing energy model 記錄控制後果。
- *service window*（服務窗口）：source = changing-service-window trace；purpose = 定義可傳輸區間；unit = trace time；



P002 | Action 的三條能量路徑

- *action*（策略動作）：source = policy API；purpose = 每次決策的唯一輸出；unit = enum；interpretation = 觸發 runner 的狀態與封包轉移。
- *WAIT*（清醒等待）：source = policy API；purpose = 保持無線端點可反應；unit = enum；interpretation = 累積清醒閒置能量。
- *SLEEP*（低功耗休息）：source = policy API；purpose = 降低休息功率；unit = enum；interpretation = 產生喚醒延遲與喚醒能量。
- *SEND_ONE*（單筆傳輸）、*SEND_URGENT*（緊急傳輸）、*FLUSH_BATCH*（批次送出）：

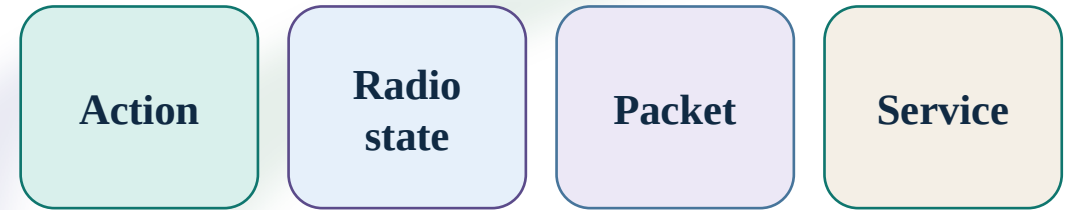
清醒等待

低功耗休息

送出封包

P003 | Action 連接狀態、封包與服務

- *radio state*（無線狀態）：source = runner event ledger；purpose = 表示休眠、清醒閒置、處理、發射與接收；unit = state enum + duration；interpretation = 提供端點能量的時間分段。



- *packet progress*（封包進度）：source = result / replay artifact；purpose = 記錄產生、佇列、嘗試、重試、送達與逾期；unit = event count / time；interpretation = 連接 action 與服務結果。

- *service result*（服務結果）：source = scenario deadline / freshness rule；purpose = 判定資料是否在規則內完成；unit = pass / fail + reason；interpretation = energy comparison

的邊界條件。

P004 | 固定情境中的改、跑、讀、解釋

• *policy*（策略）：source = marked block in *student_policy.py*；purpose = 依觀測輸入回傳合法 action；unit = code contract；interpretation = 唯一受控決策面。

• *JSON*（結構化資料格式）：source = package runner；purpose = 保存 machine-readable result；unit = data format；interpretation = 讓 result、replay 與 workbook 可核對。

• *result.json*（結果資料檔）：source = package runner；purpose = 保存 run identity、state、packet、service、energy；unit = JSON artifact；interpretation = 後續 replay / import 的輸入。

改一個值

執行 case

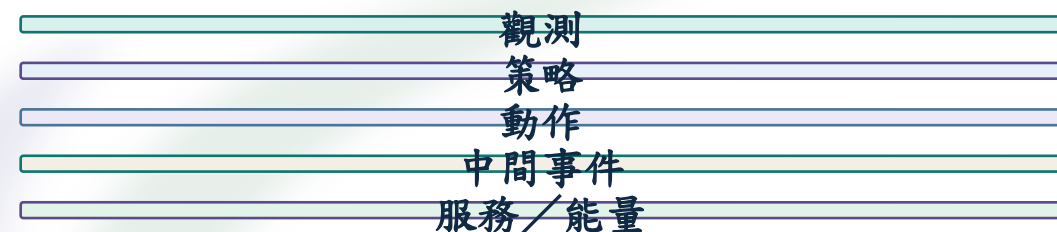
policy → evidence

解釋差異

讀取事件

P005 | 因果鏈的讀法：中間事件決定歸因

- *observation*（觀測輸入）：source = runner step；purpose = 提供目前窗口、品質、佇列與期限；unit = schema fields；interpretation = policy 可使用的當下資訊。
- *policy*（策略）與 *action*（動作）：source = student_policy.py + policy API；purpose = 由觀測選出合法控制；unit = code / enum；interpretation = 變更來源。
- *state*（狀態）與 *packet*（封包）：source = endpoint replay；purpose = 呈現狀態區間與傳輸事件；unit = duration / count / time；interpretation = 中間機制。
- *service*（服務判定）與 *endpoint energy*（端點能量）：source = scenario rule + endpoint



P006 | 可重跑 Runner 的證據範圍

- *runner*（執行器）：source = course package；purpose = 以固定 scenario、policy 與 seed 產生可重跑 artifact；unit = executable package；interpretation = controlled teaching evidence。
- *seed*（固定隨機種子）：source = scenario contract；purpose = 固定隨機序列；unit = integer identifier；interpretation = 相同輸入可重現事件順序。
- *endpoint_energy_j*（端點能量）：source = endpoint radio / processing model；purpose = 累積狀態能量；unit = J；interpretation = endpoint evidence layer。
- *coherent simulated result*（連貫模擬結果）：

固定 scenario

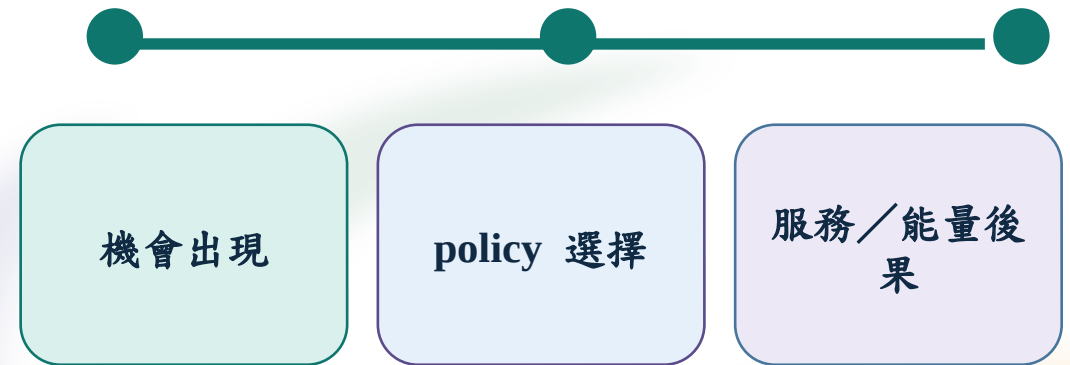
固定 seed

事件 ledger

endpoint J

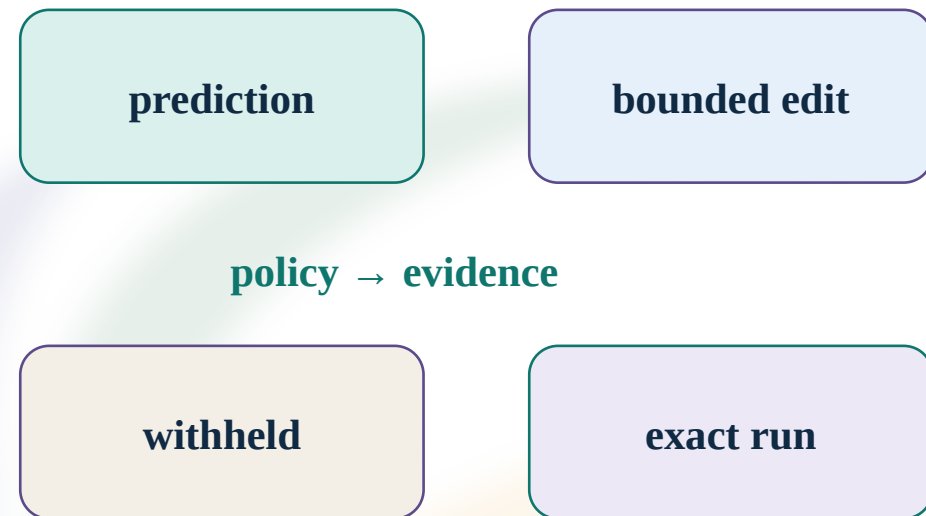
P007 | LEO 作為變動服務窗口範例

- *LEO*（低地球軌道）：source = fixed course scenario trace；purpose = 提供可開、可關、可變品質的 service-window input；unit = scenario example；interpretation = changing opportunity example。
- LEO 的作用是改變服務機會；學習焦點是端點 policy 如何在窗口變化下選擇傳輸、等待或休息。
- *HVAC*（暖通空調）：source = transfer example；purpose = 代表低負載或可連線時段；unit = application context；interpretation = 可轉移的 changing-window 機制。
- *edge gateway*（邊緣閘道）與物流運輸網路用同一機制；各情境重新定義窗口、資料期



P008 | 預測到結果：保留可反駁路徑

- *prediction*（預測）：source = workbook record；purpose = 指定 state、packet、service 或 endpoint energy 的可反駁方向；unit = statement；interpretation = run 前的比較基準。
- *workbook*（決策工作簿）：source = course route；purpose = 保存 prediction、run、replay 與 interpretation；unit = structured record；interpretation = 可重開的證據脈絡。
- *withheld case*（保留條件案例）：source = frozen scenario package；purpose = 以未調整的 policy 檢驗轉移性；unit = case label；interpretation = counterexample evidence。



P009 | Provenance 與 Claim Ceiling

- *provenance*（來源鏈）：source = package receipt、scenario、policy identity、run artifact；purpose = 連結輸入與輸出；unit = structured record；interpretation = 追溯 evidence scope。
- *source_mode*（來源模式）：source = result metadata；purpose = 區分 *student-run* 與 *same-scenario-fallback*；unit = enum；interpretation = 來源分類保持原值。
- *claim_boundary*（主張邊界）：source = course contract；purpose = 標示 evidence class；unit = fixed string；interpretation = *SIMULATED TEACHING DATA / NOT LIVE / NOT MEASURED / NOT CANONICAL-PARITY-VERIFIED*。

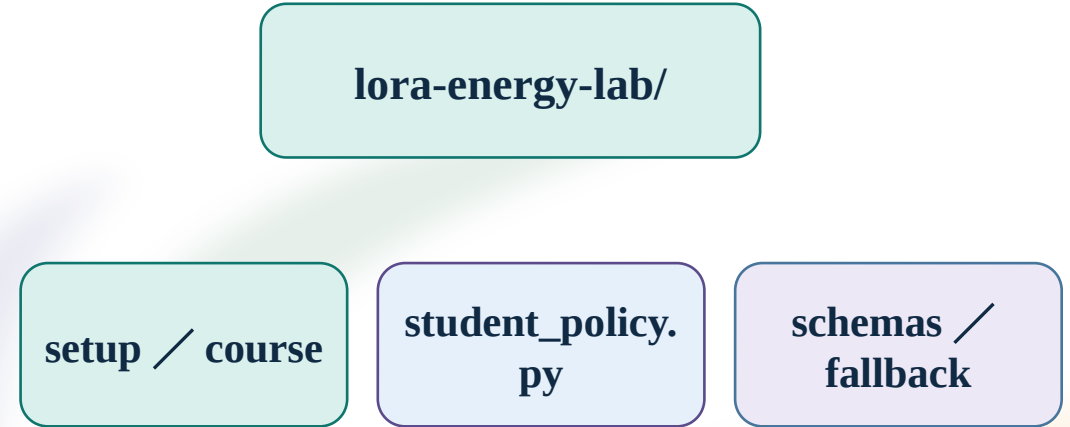
來源鏈

主張邊界

可比較欄位

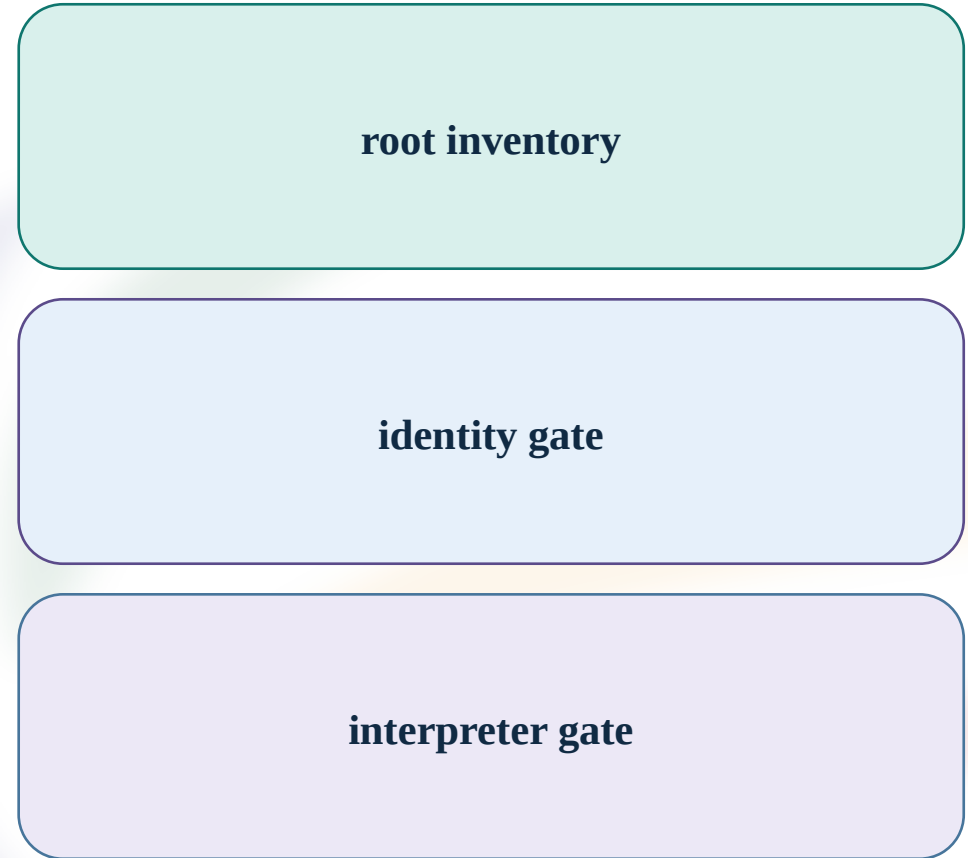
P010 | Release asset 與 package root

- *release asset*（發布封裝）：source = course package distribution；purpose = 提供同一組 runner files；unit = archive file；interpretation = package identity boundary。
- *lora-energy-lab-v1.zip*（課程封裝檔）：source = release page；purpose = transport package root；unit = file；interpretation = 解壓後形成單一 *lora-energy-lab/* 根目錄。
- *README.zh-TW.md*、*setup.sh*、*setup.cmd*、*course.sh*、*course.cmd*、*student_policy.py*：source = package root；purpose = 說明、設定、驗證、執行與 policy；unit = files；interpretation = required surface。



P011 | Package identity gate 與 root inventory

- *package identity* (套件識別) : source = release manifest + root inventory ; purpose = verify required files are one coherent package ; unit = identifier/list ; interpretation = identity mismatch stops setup 。
- *package root* (套件根目錄) : source = extracted release asset ; purpose = anchor setup.sh / setup.cmd and course.sh / course.cmd ; unit = directory ; interpretation = .venv and artifacts stay local 。
- Required inventory : README 、兩套 setup / launcher 、 *student_policy.py* 、 schemas 、 fall back_artifacts 。



P012 | Windows shell identity 與 Python 3.11

- *CMD* (Windows 命令提示字元) : source = Windows native shell ; purpose = execute *.cmd* launcher ; unit = shell type ; interpretation = path uses \ 。
- *PowerShell* (Windows 管理命令殼) : source = Windows native shell ; purpose = execute *.cmd* and inspect receipt ; unit = shell type ; interpretation = command syntax differs from CMD 。
- *Python 3.11.x* (固定直譯器版本) : source = runner frozen contract ; purpose = create package-local *.venv* ; unit = version ; interpretation = version gate before setup 。
- Commands : *py --list* ; *py -3.11 --version* →



P013 | Windows interpreter recovery

- *uv* (Python 執行環境管理工具) : source = official runtime manager ; purpose = obtain and locate exact Python 3.11 ; unit = tool / version / path ; interpretation = interpreter recovery only 。
- *uv python install 3.11* : source = uv tool ; purpose = install exact minor ; unit = command ; interpretation = creates available interpreter 。
- *uv python find 3.11* : source = uv tool ; purpose = return interpreter path ; unit = path string ; interpretation = path passes to setup 。
- Recovery result is an interpreter path ; package setup and *READY* receipt remain later gates 。

取得 uv

安裝 3.11

定位 path

進入 setup

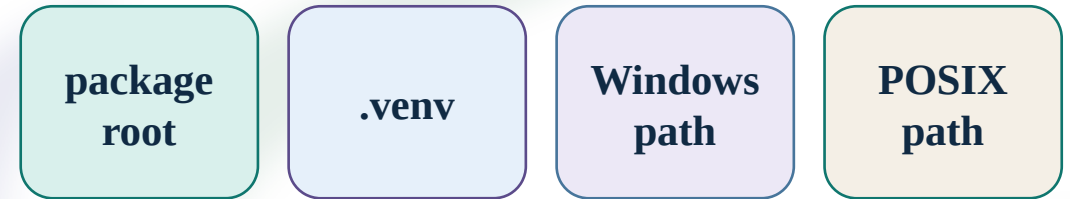
P014 | POSIX / WSL shell separation

- *POSIX* (Unix-like shell 環境) : source = macOS / Linux / WSL ; purpose = run *.sh* ; unit = environment class ; interpretation = path uses / 。
- *WSL* (Windows Subsystem for Linux ; Windows 的 Linux 執行層) : source = host platform ; purpose = run Ubuntu / Linux tools ; unit = environment class ; interpretation = owns a separate *.venv* 。
- *Ubuntu* (Linux 發行版) : source = WSL distribution ; purpose = provide verified Linux shell ; unit = distribution name ; interpretation = *setup.sh* and *course.sh* run inside this boundary 。



P015 | package-local .venv 隔離

- `.venv`（套件本地虛擬環境）：source = Python `venv` module；purpose = isolate locked dependencies and policy；unit = directory；interpretation = launcher uses its own interpreter。
- `venv`（虛擬環境標準模組）：source = Python standard library；purpose = create `.venv`；unit = module；interpretation = environment creation mechanism, not an interpreter version。
- Windows path：`.venv\Scripts\python.exe`；
POSIX path：`.venv/bin/python`。
- Package-local Python should report Python 3.11.x；啟用只作診斷，正常 launcher 直接



同一 platform boundary

P016 | Windows setup 與 READY receipt

- *setup.cmd* (Windows 設定啟動器) : source = package root ; purpose = create .venv, install locked dependencies, run verification ; unit = executable file ; interpretation = prepares package boundary 。
- *course.cmd verify* (Windows 驗證命令) : source = package launcher ; purpose = re-check interpreter / lock / scenario / policy contract ; unit = command ; interpretation = emits verification receipt 。
- *READY* (環境就緒狀態) : source = *artifacts/verify-receipt.json* ; purpose = indicate environment / contract gate passed ; unit = enum ; interpretation = runner may enter case execution 。
- Expected fields : status READY 、 Python 3.11.x 、 scenario identity 、 engine mode 、 claim boundary 。

setup.cmd

course.cmd verify

READY

verify receipt

P017 | WSL identity gate

- `wsl --install -d Ubuntu` (WSL 安裝命令) :
source = Windows PowerShell ; purpose =
create Ubuntu distribution when absent ; unit =
command ; interpretation = host setup only 。
- `wsl -l -v` (WSL 分發清單) : source =
Windows command ; purpose = show
distribution / version / status ; unit = list ;
interpretation = selects correct Ubuntu
boundary 。
- `wsl -d Ubuntu` (進入命令) : source =
Windows command ; purpose = open Ubuntu
shell ; unit = command ; interpretation =
subsequent `.sh` commands belong to Linux 。
- Ubuntu shell evidence : `pwd` Linux path and

Windows shell

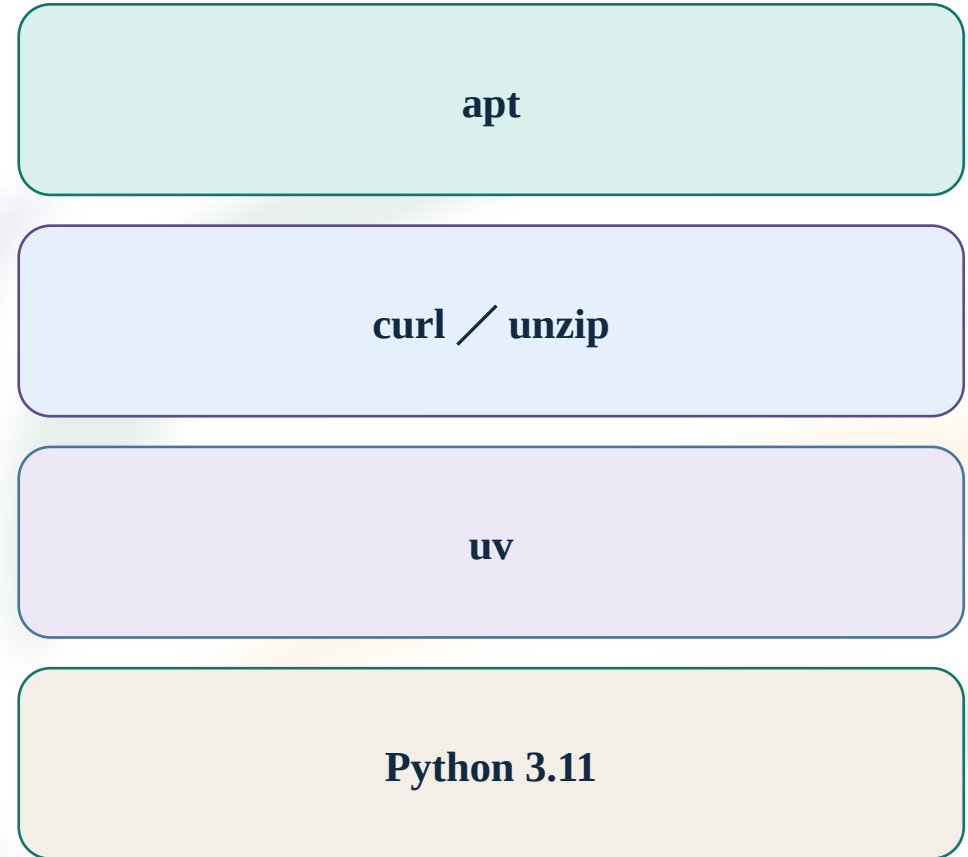
`wsl -l -v`

Ubuntu

Linux shell

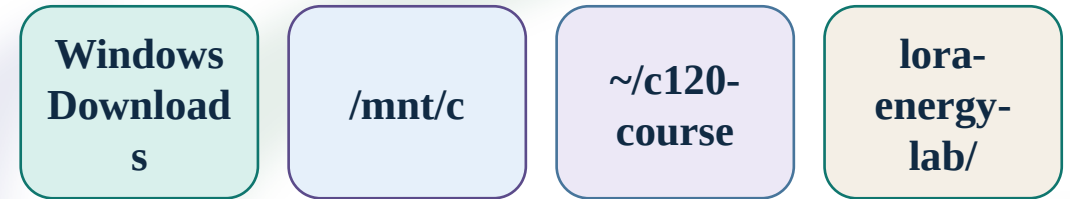
P018 | WSL toolchain ladder

- *apt* (Linux 套件管理器) : source = Ubuntu ; purpose = obtain *curl* and *unzip* ; unit = command / tool ; interpretation = provides release extraction prerequisites 。
- *curl* (命令列傳輸工具) : source = Ubuntu package ; purpose = retrieve uv installer ; unit = tool ; interpretation = environment setup input 。
- *unzip* (封裝解壓工具) : source = Ubuntu package ; purpose = expand release asset ; unit = tool ; interpretation = creates package root 。
- Order : apt update / install uv version / install / find → Python 3.11 path 。



P019 | WSL package path

- `/mnt/c/.../Downloads` (Windows mount path) : source = WSL mount ; purpose = read release asset from Windows Downloads ; unit = filesystem path ; interpretation = transfer source only 。
- `~/c120-course` (Linux course staging path) : source = Ubuntu home ; purpose = hold release files and extracted root ; unit = directory ; interpretation = Linux-owned `.venv` boundary 。
- `cp` (複製命令) : source = POSIX shell ; purpose = copy release files into Linux home ; unit = command ; interpretation = prevents cross-platform environment mixing 。
- `pwd` (目前目錄命令) : source = POSIX



同一 platform boundary

P020 | POSIX / WSL setup 與 verify

- *setup.sh* (POSIX 設定啟動器) : source = package root ; purpose = create *.venv*, install lock dependencies, call verification ; unit = executable file ; interpretation = prepares Linux / macOS / WSL runner 。
- *course.sh verify* (POSIX 驗證命令) : source = package launcher ; purpose = check exact interpreter and course contract ; unit = command ; interpretation = writes *artifacts/verify-receipt.json* 。
- *PYTHON_BIN* (Python 執行器路徑變數) : source = shell environment ; purpose = select *python3.11* or *uv python find 3.11* output ; unit = path string ; interpretation = setup input 。
- Expected status : READY ; Python 3.11.x ; package-local *.venv/bin/python* 。

PYTHON_BIN

setup.sh

course.sh verify

READY

P021a | 查看 READY receipt

- *verify-receipt.json* (驗證收據檔) : source = setup / verify command ; purpose = record environment and contract gate ; unit = JSON artifact ; interpretation = setup evidence 。
- POSIX : *sed -n '1,80p' artifacts/verify-receipt.json*
- Windows PowerShell : *Get-Content .\artifacts\verify-receipt.json*
- Windows CMD : *type artifacts/verify-receipt.json*
- Read fields as structured data ; preserve file provenance 。

POSIX

PowerShell

CMD

receipt

P021b | READY receipt : 狀態與識別欄位

ready

- *status* (狀態) : source = receipt ; purpose = gate result ; unit = enum ; interpretation = READY permits case entry 。
- *python_version* (Python 版本) : source = receipt ; purpose = record interpreter ; unit = version ; interpretation = must be 3.11.x 。
- *scenario_id* (情境識別碼) : source = scenario contract / receipt ; purpose = bind fixed case ; unit = string identifier ; interpretation = identifies the scenario, not an energy result 。
- *policy_api_version* (策略介面版本) : source = package contract ; purpose = bind allowed observations / actions ; unit = string identifier ; interpretation = prevents API drift 。

P021c | READY receipt : 引擎與主張欄位

- *engine_mode* (執行引擎模式) : source = runner metadata ; purpose = classify producer ; unit = enum ; interpretation = *coherent-course-simulated-adapter* 。
- *upstream_execution* (上游程式執行狀態) : source = runner metadata ; purpose = state whether upstream repo ran ; unit = Boolean ; interpretation = false for course adapter 。
- *claim_boundary* (主張邊界) : source = course contract ; purpose = label evidence ; unit = fixed string ; interpretation = *SIMULATED TEACHING DATA / NOT LIVE / NOT MEASURED / NOT CANONICAL-PARITY-VERIFIED* 。
- Receipt 通過表示環境與契約一致 ; case result 仍需另行產生並以自己的 run identity 保存 。

P022 | Setup gate recovery 與 same-scenario fallback

- *artifact_source*（資料來源分類）：source = result metadata；purpose = label local policy run or fallback；unit = enum；interpretation = source mode remains visible。
- *same-scenario-fallback*（同情境回復資料）：source = package *fallback_artifacts/*；purpose = continue a case when local setup is blocked；unit = artifact pair；interpretation = same scenario identity, no local policy execution claim。
- *fallback_artifacts/<case>/result.json*（回復結果檔）與 *endpoint-replay.json*（端點重播檔）：source = package artifact；purpose = preserve result / replay pairing；unit = JSON files；interpretation = imported together。
- Recovery decision：修復指示的 boundary 或使
用 exact case pair；保留錯誤與來源分類。

環境 gate

修復 boundary

same-scenario-
fallback

P023 | Bounded edit surface : student_policy.py

- *student_policy.py* (策略檔名) : source = package root ; purpose = expose three marked blocks ; unit = Python file ; interpretation = only editable course surface 。
- *marked block* (標記區段) : source = file markers ; purpose = bound one lab edit ; unit = source range ; interpretation = keeps policy identity and result lineage interpretable 。
- *lab-a-pace-rest* 、 *lab-b-enter-exit-hold* 、 *lab-c-batch-urgent* : source = marked block labels ; purpose = select current lab control ; unit = identifier ; interpretation = one active region per edit 。
- Scenario 、 schemas 、 runner 、 網站程式與 generated artifact remain read-only package inputs / outputs 。

student_policy.py

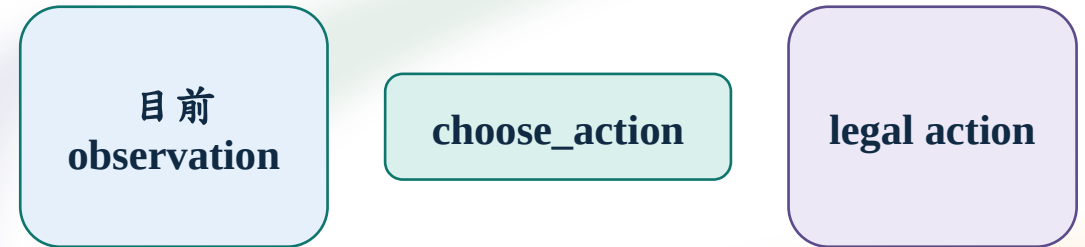
A marked block

B marked block

C marked block

P024a | Observation 是 decision-time input

- *choose_action(observation)* (策略函式) :
source = policy API ; purpose = read current observation and return one legal action ; unit = function contract ; interpretation = no future / result access 。
- *contact_open* (窗口開啟狀態) : source = current scenario observation ; purpose = guard send legality ; unit = Boolean ; interpretation = false routes to SLEEP 。
- *quality_band* (品質分帶) : source = current trace observation ; purpose = expose present link quality category ; unit = ordinal band ; interpretation = threshold input 。
- *stable_steps* (穩定步數) : source = trace



P024b | Observation 欄位與合法 action

- *steps_since_send* （距上次送出步數）：
source = runner state ； purpose = pacing
input ； unit = steps ； interpretation = guards
rest gap 。
- *queue_size* （佇列數量）： source = endpoint
queue ； purpose = batching / urgency input ；
unit = packet count ； interpretation =
determines send pressure 。
- *urgent_due_in_s* （緊急期限剩餘秒數）：
source = scenario deadline ； purpose = urgency
input ； unit = 秒（s）； interpretation =
compares with urgency margin 。
- *previous_action* （前一動作）： source =
runner state ； purpose = detect transitions ；

queue /
deadline

contact / pace

合法輸出

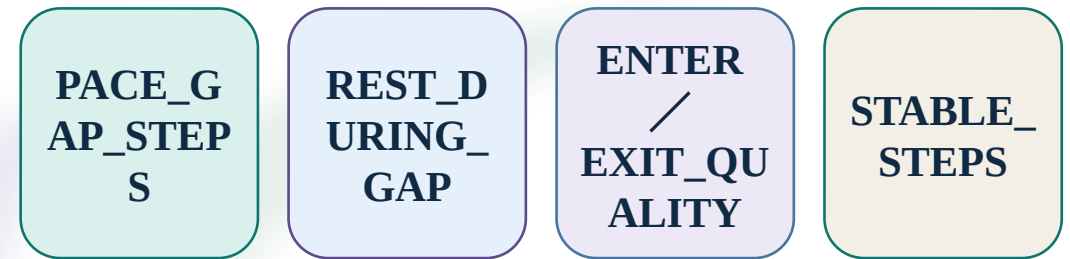
P025a | Lab A / B 的受控常數

• *PACE_GAP_STEPS*（節奏間隔步數）：source = Lab A marked block；purpose = minimum steps between sends；unit = steps；interpretation = pacing boundary。

• *REST_DURING_GAP*（空檔休息動作）：source = Lab A marked block；purpose = choose WAIT or SLEEP during gap；unit = action enum；interpretation = changes awake-idle versus wake cost。

• *ENTER_QUALITY*（進入品質門檻）：source = Lab B marked block；purpose = activate send mode；unit = quality band；interpretation = enter threshold。

• *EXIT_QUALITY*（退出品質門檻）：source = Lab B marked block；purpose = deactivate send mode；unit = quality band；interpretation = exit threshold。



一次一個受控常數

P025b | Lab C 的佇列與緊急常數

- **BATCH_SIZE**（批次大小）：source = Lab C marked block；purpose = set packets grouped for flush；unit = packet count；interpretation = controls batching and wait energy。
- **URGENT_MARGIN_S**（緊急餘裕秒數）：source = Lab C marked block；purpose = trigger urgent action before deadline；unit = 秒（s）；interpretation = changes deadline protection。
- **FLUSH_BATCH**（批次送出）：source = policy API；purpose = transmit queued packets as a group；unit = action enum；interpretation = links queue to process / TX / RX and delivery。
- Candidate、revision 各改一個 marked constant；withheld surprise 維持 frozen policy。



一次一個受控常數

P026a | 讀取條件與 action 分支

- Policy branch order is evaluated from current observation 。
- $contact_open = false \rightarrow SLEEP$: source = policy API ; purpose = keep send illegal outside window ; unit = Boolean $\rightarrow$ enum ; interpretation = protects service legality 。
- $urgent_pending$ (緊急封包待處理) : source = queue / deadline observation ; purpose = reveal urgent packet ; unit = Boolean ; interpretation = activates deadline branch 。
- $urgent_due_in_s \leq URGENT_MARGIN_S \rightarrow SEND_URGENT$: source = deadline + policy constant ; purpose = protect urgent deadline ; unit = 秒比較 ; interpretation = earlier trigger as margin grows 。
- $steps_since_send < PACE_GAP_STEPS \rightarrow REST_DURING_GAP$: source = pacing observation



P026b | Action 後果的事件讀法

- *WAIT* → 清醒閒置 interval ; source = runner ledger ; purpose = retain response ; unit = duration / J ; interpretation = time-integrated energy cost 。
- *SLEEP* → 休眠 interval + wake transition ; source = runner ledger ; purpose = reduce idle power ; unit = duration / J ; interpretation = wake latency / energy appear in event chain 。
- *SEND_** → process / TX / RX / attempt / retry / delivery ; source = endpoint replay ; purpose = show packet path ; unit = duration / count ; interpretation = connects action to service 。



- *endpoint_energy_j* 與 service result 一起判讀；

P027a | Python indentation 與 return

- *indentation* (縮排) : source = Python syntax ; purpose = assign statements to branch ; unit = code structure ; interpretation = determines branch membership 。
- *return* (回傳) : source = policy function ; purpose = send one legal action to runner ; unit = action enum ; interpretation = closes decision step 。
- Minimal skeleton : *def choose_action(observation):* → *if not observation.contact_open:* → *return SLEEP* 。
- Syntax gate failure is repaired inside the active marked block ; runner 、 scenario schema 維持原狀 。

observation
↓ branch

return action

P027b | Lab A baseline run 與結果入口

- *baseline*（基線執行）：source = packaged policy + fixed scenario / case；purpose = provide control artifact；unit = case label；interpretation = comparison anchor。
- *bash course.sh run --lab A --case baseline / course.cmd run --lab A --case baseline*：source = package launcher；purpose = execute exact case；unit = command；interpretation = emits result path and run identity。
- *result_path*（結果檔路徑）：source = command JSON output；purpose = locate *result.json*；unit = path string；interpretation = import and replay pair entry。
- *endpoint-replay.json*（端點重播檔）：source

baseline

result.json

endpoint-replay.json

run identity